

Dimension Reduction in Statistics and Machine Learning

Lecture 8 – Introduction to Modern Machine Learning

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Outline

1. Why dimension reduction?
2. Principal component analysis
3. Examples and interpretation
4. Beyond PCA
5. Putting it into practice

Why dimension reduction?

Where this lecture fits

Earlier lectures introduced:

- supervised learning, k -nearest neighbours, regression and classification;
- loss functions, generalisation, cross-validation and regularisation;
- decision trees, bagging and random forests;
- online learning.

Today

We focus on **representations**: replacing many original features by a smaller number of informative coordinates.

This is usually an **unsupervised learning** task, but it often sits inside a supervised ML pipeline.

What is dimension reduction?

Suppose each observation is a vector

$$x^{(i)} = (x_1^{(i)}, \dots, x_p^{(i)}) \in \mathbb{R}^p.$$

Dimension reduction

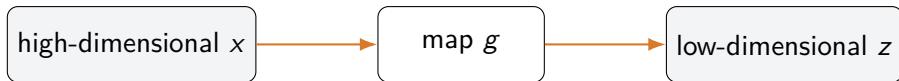
Find a map

$$g : \mathbb{R}^p \longrightarrow \mathbb{R}^k, \quad k \ll p,$$

so that the low-dimensional representation

$$z^{(i)} = g(x^{(i)}) \in \mathbb{R}^k$$

preserves the information needed for the task.



Data matrix notation

Collect the data in a matrix

$$X = \begin{bmatrix} - & (x^{(1)})^\top & - \\ - & (x^{(2)})^\top & - \\ & \vdots & \\ - & (x^{(N)})^\top & - \end{bmatrix} \in \mathbb{R}^{N \times p}.$$

- Rows = observations.
- Columns = features.
- Dimension reduction replaces X by a score matrix

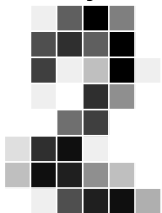
$$Z \in \mathbb{R}^{N \times k}, \quad k \ll p.$$

Key question

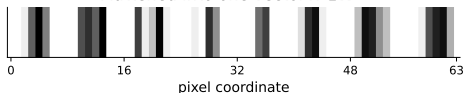
What should “preserve information” mean? It depends on the goal: visualisation, compression, denoising, prediction, clustering, or interpretation.

Running example: images as vectors

An 8×8 image of a digit



Flattened into one vector $x \in \mathbb{R}^{64}$



A greyscale image can be treated as a vector:

$$8 \times 8 \text{ pixels} \longrightarrow x \in \mathbb{R}^{64}.$$

- Each coordinate is a pixel intensity.
- A dataset of images is a cloud of points in a high-dimensional feature space.
- We will use handwritten digits as a running example.

Why reduce dimension?

Statistical reasons

- Reduce noise.
- Stabilise estimates when p is large.
- Discover latent structure.
- Improve interpretability.

Computational/ML reasons

- Visualise data in 2D or 3D.
- Compress images or signals.
- Speed up downstream algorithms.
- Build features for supervised learning.

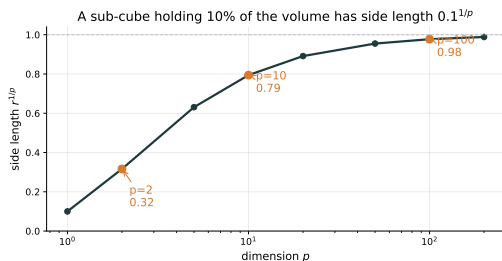
Important

Dimension reduction is not automatically harmless: it can discard information, especially if the objective of the reduction is not aligned with the final task.

Nothing is local in high dimensions

Suppose points are spread over the unit cube $[0, 1]^p$. A smaller cube that captures fraction r of the volume has side length

$$\ell = r^{1/p}.$$



For $r = 0.10$:

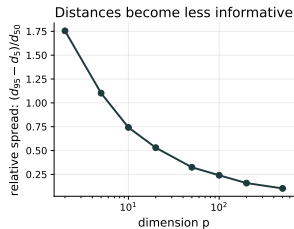
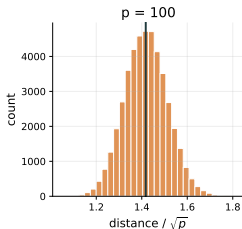
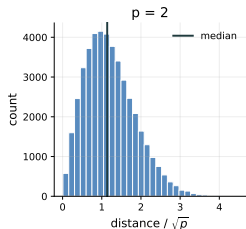
p	2	10	100
$0.1^{1/p}$	0.32	0.79	0.98

Connection to kNN

A “neighbourhood” containing only 10% of the data can span almost the whole range of each coordinate.

High-dimensional geometry can be strange

Points sampled independently from $N(0, I_p)$:



Distance concentration

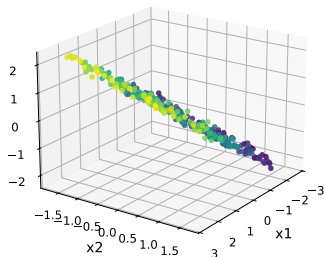
For random points in high dimensions, pairwise distances often become similar.

- Nearest neighbours may not be much nearer than farthest neighbours.
- Visual intuition from $\mathbb{R}^2$ and $\mathbb{R}^3$ can fail.

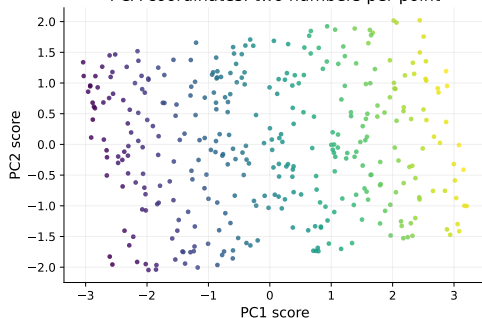
The optimistic view: low-dimensional structure

High-dimensional data are often not arbitrary points in $\mathbb{R}^p$. They may be well described by a smaller number of latent factors.

Data live near a 2D plane in 3D



PCA coordinates: two numbers per point



A 3D point cloud near a 2D plane can be represented with two PCA coordinates.

Principal component analysis

PCA in one sentence

Principal component analysis

PCA finds orthonormal directions in feature space along which the centered data have maximal variance.

Input:

- (centered) data matrix
 $X \in \mathbb{R}^{N \times p}$;
- choose number of principal components k .

Output:

- loading vectors
 $v_1, \dots, v_k \in \mathbb{R}^p$;
- scores $z^{(i)} = ((x^{(i)})^\top v_1, \dots, (x^{(i)})^\top v_k)$.

PCA is a linear method: it represents data as linear combinations of the original features, which geometrically amounts to rotating the axes and projecting onto them.

How PCA is computed: an animated view

Maximise projected variance $\Rightarrow$ loading directions v_j ; project observations onto them $\Rightarrow$ score coordinates z_i .

Step 0: center the data

Let X be a data matrix and

$$\bar{x} = \frac{1}{N} \sum_{i=1}^N x^{(i)} \in \mathbb{R}^p.$$

Define the centered observations

$$x_c^{(i)} = x^{(i)} - \bar{x},$$

and the centered data matrix

$$X_c = X - \mathbf{1}\bar{x}^\top.$$

Why center?

PCA is about directions of variation around the mean, not about the position of the cloud relative to the origin.

If variables are measured on very different scales, standardise columns before PCA unless the scale itself is meaningful.

The first principal component

Let the sample covariance matrix be

$$S = \frac{1}{N-1} X_c^\top X_c \in \mathbb{R}^{p \times p}.$$

For a unit vector v , the projected coordinate of observation i is

$$z^{(i)} = (x_c^{(i)})^\top v.$$

The sample variance of the projected data is

$$\frac{1}{N-1} \sum_{i=1}^N \left((x_c^{(i)})^\top v \right)^2 = v^\top S v.$$

Optimisation problem

$$v_1 = \arg \max_{\|v\|_2=1} v^\top S v.$$

Solving the PCA optimisation problem

Since S is symmetric and positive semi-definite, it has eigenvectors

$$Sv_j = \lambda_j v_j, \quad \lambda_1 \geq \lambda_2 \geq \dots \geq 0.$$

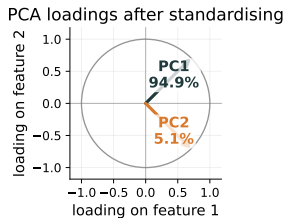
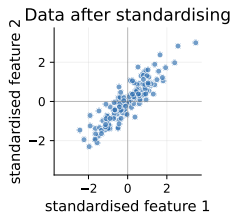
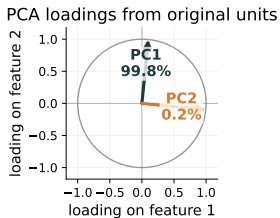
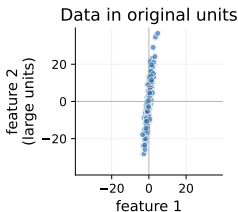
- v_1 is the unit eigenvector with largest eigenvalue λ_1 .
- The variance explained by PC j is λ_j .
- Subsequent PCs solve the same problem subject to orthonormality:

$$v_j \perp v_1, \dots, v_{j-1}, \quad \|v_j\|_2 = 1.$$

Sign convention

If v_j is a principal component, then $-v_j$ is also a principal component. The sign is arbitrary.

Scaling can change PCA



Rule of thumb

PCA on the covariance matrix is sensitive to feature units.

From p features to k scores

Let

$$V_k = [v_1 \ v_2 \ \cdots \ v_k] \in \mathbb{R}^{p \times k}, \quad V_k^\top V_k = I_k.$$

The PCA scores are

$$Z = X_c V_k \in \mathbb{R}^{N \times k}.$$

The reconstruction from the rank- k approximation is

$$\hat{X}_k = Z V_k^\top + \mathbf{1} \bar{x}^\top = X_c V_k V_k^\top + \mathbf{1} \bar{x}^\top.$$

Geometric interpretation

$X_c V_k V_k^\top$ is the orthogonal projection of the centered data onto the k -dimensional subspace spanned by the first k principal components.

Loadings vs scores: do not confuse them

Loadings / principal directions

$$V_k = [v_1, \dots, v_k] \in \mathbb{R}^{p \times k}$$

- Each v_j is a direction in the original **feature space**.
- It has p entries: one value per original feature.
- It tells us how variables combine to define a component.

Scores / new coordinates

$$Z = X_c V_k \in \mathbb{R}^{N \times k}$$

- Row i gives the coordinates of observation i in the PCA system.
- It has k entries: one coordinate per retained component.
- It tells us where the observation sits in the lower-dimensional space.

Common confusion

PC1 is a vector in $\mathbb{R}^p$; a PC1 score is the scalar $(x_c^{(i)})^\top v_1$.

PCA as optimal reconstruction

PCA has a second, equivalent interpretation:¹

$$V_k = \arg \min_{V^T V = I_k} \|X_c - X_c V V^T\|_F^2.$$

- It chooses the k -dimensional linear subspace with the smallest squared reconstruction error.
- The fraction of total variance explained by the first k PCs is

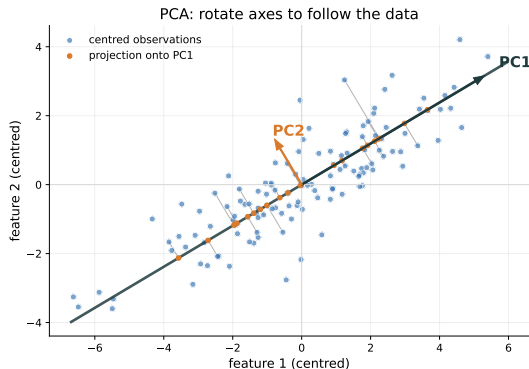
$$\text{PVE}(k) = \frac{\lambda_1 + \dots + \lambda_k}{\lambda_1 + \dots + \lambda_p}.$$

Takeaway

Maximising variance and minimising reconstruction error lead to the same PCA subspace.

¹Here $\|A\|_F^2 = \sum_{i,j} A_{ij}^2 = \text{Trace}(A^T A)$ denotes the squared Frobenius norm.

PCA geometry



- PC1 follows the longest direction of the point cloud.
- PC2 is orthogonal to PC1.
- Projecting onto PC1 replaces each point by one coordinate.

Question

What information is lost when we keep only PC1?

Computing PCA using the SVD

Computing eigenvectors of S can be expensive when p is large. The singular value decomposition is often the most stable route.

For centered data:²

$$X_c = UDV^\top,$$

where $D = \text{diag}(d_1, \dots, d_r)$ with $d_1 \geq \dots \geq d_r > 0$ and $r = \text{rank}(X_c)$.

Loadings

Columns of V are the principal directions:

$$v_1, v_2, \dots$$

Scores

$$Z = X_c V_k = U_k D_k.$$

The covariance eigenvalues are $\lambda_j = d_j^2 / (N - 1)$.

²Write U_k, V_k for the first k columns of U, V , and $D_k = \text{diag}(d_1, \dots, d_k)$.

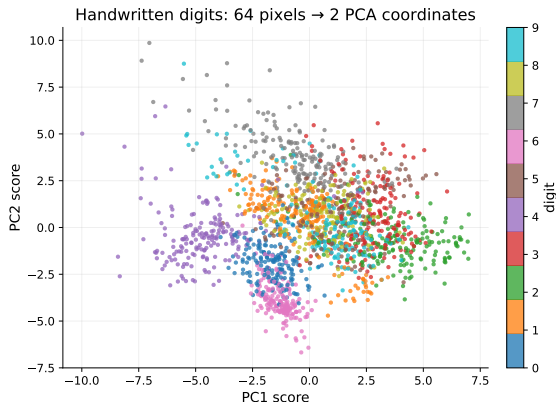
A practical PCA algorithm

- ① Put data into a matrix $X \in \mathbb{R}^{N \times p}$.
- ② Center each column; standardise columns if needed.
- ③ Compute PCA using eigendecomposition of S or SVD of X_c .
- ④ Choose the number of components k .
- ⑤ Use $Z = X_c V_k$ for visualisation, compression or as features.

Examples and interpretation

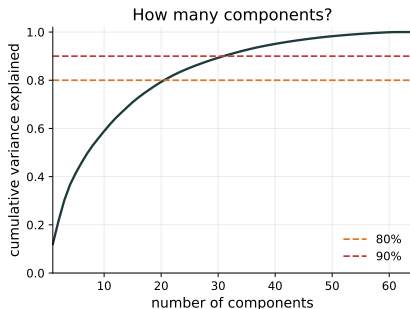
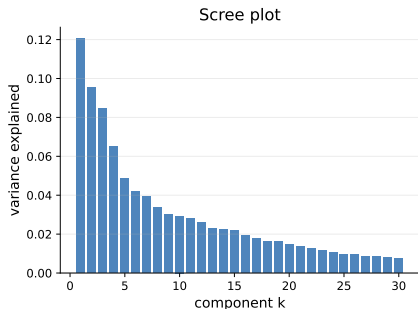
Example: handwritten digits

Each image is 8×8 , so each observation has $p = 64$ pixel features. PCA gives a two-dimensional representation.



- Nearby points tend to correspond to visually similar digits.
- The two-dimensional plot is useful, but it does not perfectly separate all classes.
- This is exploratory: labels were not used by PCA.

Explained variance: a scree plot



How to read this

The first few components explain the largest individual shares of variance. The cumulative curve shows how many components are needed to retain a chosen fraction of total variance.

How should we choose k ?

There is no universal rule. Common approaches:

- **Visualisation:** choose $k = 2$ or 3 .
- **Compression:** choose the smallest k with acceptable reconstruction error.
- **Explained variance:** choose k so that $\text{PVE}(k)$ exceeds a threshold such as 80% or 90%.
- **Prediction:** choose k using validation performance inside the ML pipeline.
- **Interpretability:** choose a small k whose loadings can be explained.

Caution

High explained variance does not guarantee high predictive accuracy. A direction can explain little variance overall and still contain important label information.

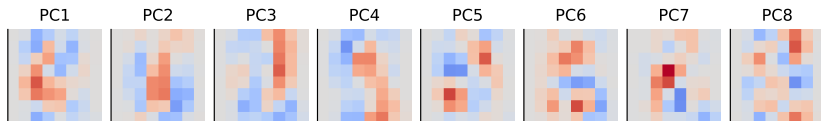
PCA reconstruction: compression (animated)

- With small k , the reconstruction is blurry but often recognisable.
- Increasing k keeps more detail and raises cumulative explained variance.

Interpreting components: loadings

The loading vector $v_j \in \mathbb{R}^p$ tells us how the original features contribute to PC j .

Principal directions as 'eigendigits'



- Large positive or negative entries indicate features that strongly affect the component score.
- For images, loadings can be reshaped into “eigenimages”.
- For tabular data, loadings help explain what each component measures.

PCA inside a supervised learning pipeline

Dimension reduction often appears before a supervised model:



When can this help?

- many correlated features;
- noisy measurements;
- algorithms whose cost grows quickly with p ;
- visual diagnostics before modelling.

Data leakage: fit transformations only on training data

If PCA is used before a supervised model, it must be learned without looking at the test data.

Wrong

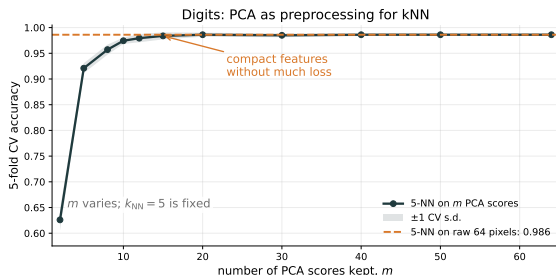
- ❶ Fit scaler and PCA on all data.
- ❷ Split into train/test.
- ❸ Evaluate model.

Right

- ❶ Split into train/test.
- ❷ Fit scaler and PCA on training data.
- ❸ Apply learned transform to test data.

During cross-validation, the scaler, PCA and any choice of k must be refit inside each training fold.

PCA as preprocessing: kNN on digits



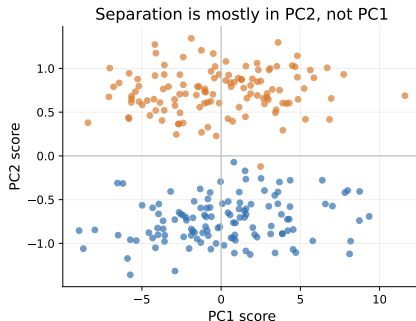
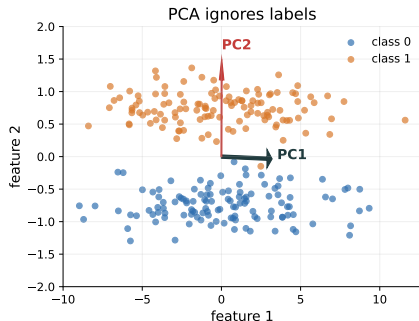
Recall k -nearest neighbours from Lecture 1: it relies directly on distances between observations.

On the digits data, PCA can replace 64 pixel coordinates by a much smaller number of scores.

Takeaway

PCA is not only for plotting. It can create compact features for later models, but hyperparameters must be validated inside the full pipeline.

What can go wrong? PCA ignores labels



Lesson

PCA finds high-variance directions, not the most predictive directions. For supervised tasks, validate the full pipeline.

Is PCA variable selection like LASSO?

LASSO / variable selection

$$\min_{\beta} \text{loss}(\beta) + \lambda \sum_{j=1}^p |\beta_j|$$

- works with a supervised target y ;
- can set some coefficients exactly to zero;
- keeps a subset of the original variables.

PCA / new coordinates

$$z_j = X_c v_j$$

- uses only the features X ;
- components are usually dense linear combinations;
- creates new variables rather than selecting old ones.

Takeaway

PCA reduces dimension by changing coordinates; LASSO reduces dimension by shrinking/selecting original features.

Limitations of PCA

- **Linear:** PCA finds a flat subspace, not a curved surface.
- **Scale-sensitive:** variables with large units can dominate.
- **Outlier-sensitive:** a few extreme observations can rotate components.
- **Unsupervised:** it may discard low-variance directions that predict labels.
- **Interpretability:** PCs are linear combinations of original variables and may be hard to understand.

Practical advice

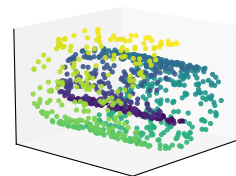
Use PCA as a baseline. Plot scores and loadings. Check reconstruction or downstream validation. Keep the preprocessing pipeline reproducible.

Beyond PCA

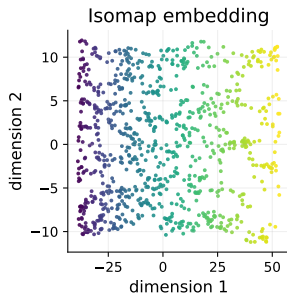
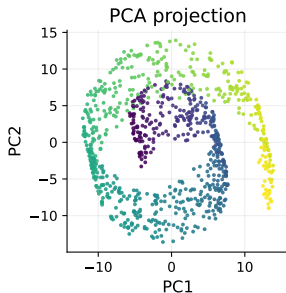
When the structure is curved

PCA can fail when the data lie near a curved low-dimensional manifold.

3D data: a rolled-up sheet



colour = roll position



- PCA projects the roll onto a flat plane and overlaps distant parts.
- Nonlinear methods try to preserve local neighbourhood or geodesic structure.

Neighbourhood-preserving embeddings

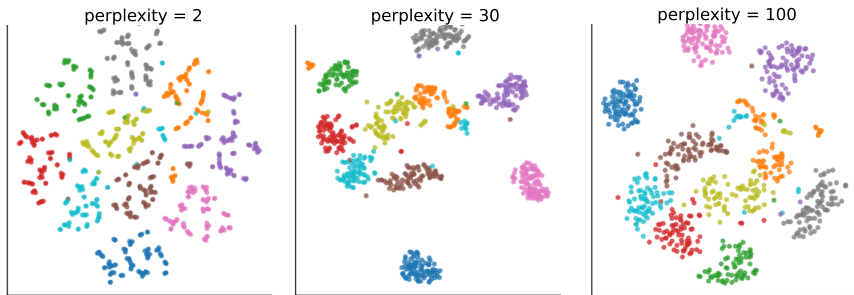
Common nonlinear methods for visualisation include:

- **Isomap**: build a neighbourhood graph, approximate geodesic distances by shortest paths, then apply MDS.
 - **MDS**: place points so embedded Euclidean distances approximate chosen pairwise dissimilarities.
- **t-SNE**: preserve local neighbourhood probabilities for 2D/3D visualisation.
- **UMAP**: construct a weighted/fuzzy nearest-neighbour graph and optimise a low-dimensional graph representation.

Interpretation warning

In t-SNE/UMAP plots, cluster sizes, distances between clusters and empty spaces can be misleading. They are also sensitive to hyperparameters. Use them as exploratory visualisations.

t-SNE: hyperparameters can change the story



Practical warning

The same data can look different under different settings. For t-SNE and UMAP, check sensitivity to hyperparameters and avoid over-interpreting distances between clusters.

Autoencoders: nonlinear dimension reduction

A neural network autoencoder learns two functions:

$$\text{encoder } e_{\theta} : \mathbb{R}^p \rightarrow \mathbb{R}^k, \quad \text{decoder } d_{\phi} : \mathbb{R}^k \rightarrow \mathbb{R}^p.$$

It is trained by minimising reconstruction loss:

$$\min_{\theta, \phi} \frac{1}{N} \sum_{i=1}^N \left\| x^{(i)} - d_{\phi}(e_{\theta}(x^{(i)})) \right\|_2^2.$$



This connects to later lectures on neural networks and generative models.

Choosing a method

Method	Main idea	Best for	Watch out for
PCA	Linear subspace; maximise variance	baseline, compression, denoising, interpretable loadings	scaling, outliers, nonlinear structure
MDS/Isomap	Preserve distances or graph distances	curved manifolds, exploratory structure	neighbourhood choices, noise
t-SNE/UMAP	Preserve local neighbourhoods in 2D/3D	visualisation of complex datasets	distances and axes are not directly interpretable
Autoencoder	Learn nonlinear encoder and decoder	large data, images/signals, flexible features	tuning, overfitting, lower interpretability

Default workflow

Start with PCA. Move to nonlinear methods only when the goal and diagnostics justify the extra complexity.

Putting it into practice

Mini-exercise: think before computing

You have $N = 500$ patients and $p = 20,000$ gene-expression measurements. You want to predict whether a patient responds to a treatment.

Discuss with a neighbour:

- ① What are two reasons dimension reduction might help?
- ② Should you standardise the variables before PCA?
- ③ If you evaluate a classifier after PCA, where does PCA sit in the cross-validation loop?
- ④ Could a component explaining little variance still be predictive?

Answer sketch

High p can increase noise and computational cost; standardisation is often needed; PCA must be fit inside each training fold; yes, low-variance directions can matter for prediction.

Summary

- Dimension reduction replaces p original features by $k \ll p$ coordinates.
- PCA finds orthogonal directions of maximum variance in centered data.
- PCA is equivalent to finding the best rank- k linear reconstruction of the centered data.
- Choose k according to the task: visualisation, compression, explained variance, validation or interpretability.
- Fit PCA, and any choice of k , inside the training data pipeline to avoid leakage.
- PCA creates new coordinates; it is not the same as selecting original variables.
- PCA is linear and unsupervised; nonlinear methods also exist and can address different goals.